The Mirage Test: An Empirical Study of LLM-as-Judge Reliability via Behavioral Invariance in the Repeated Prisoner's Dilemma

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Abstract

We study the reliability of LLM-as-judge evaluation pipelines through a natural invariance property. A judge that scores agent behavior should depend on a transcript only through the actions the agent took, not through the surface text (free-form rationale, dialogue, or styling) that accompanies those actions. We test this property on a finitely repeated Prisoner's Dilemma testbed by constructing matched pairs of transcripts, one generated by LLM agents and one generated by scripted strategies, whose joint round-by-round action sequences agree on at least sixteen of twenty player-round actions (joint Hamming distance at most four out of twenty) but whose surface content differs systematically. The judge returns a six-dimensional mental-state attribution score (beliefs, desires, intentions, metacognition, self-awareness, moral patienthood), each component an integer on a one-to-five Likert scale; we aggregate to a single mean score $\bar{J}$. On seventeen matched pairs, the gap between the judge's aggregate score on LLM transcripts and on behaviorally matched scripted transcripts is $\hat{\Delta} = +1.6471$ ($p = 2.91 \times 10^{-4}$, matched-pair Cohen's $d = +3.61$), with the gap present on every one of the six rubric dimensions under FDR correction. A six-cell ablation that holds the action sequence fixed within matched index decomposes this effect: wrapping LLM-style rationale around scripted action sequences closes essentially the entire gap from below, and removing rationale from LLM transcripts opens it from above. A factorial regression with action-feature controls gives a source-by-reasoning interaction of $\hat{\beta}_3 = -1.637$ ($p < 10^{-3}$); interpreted conditionally, the source effect falls from about $+1.80$ points at the marker-stripped level to a residual of about $+0.16$ when rationale richness is matched. A small persuasion pilot at fixed action sequence shows that prompt-engineered rationale moves $\bar{J}$ upward across three canonical specifications, though the effect is smaller than the matched-pair gap. We propose, as a working conjecture, that the judge performs in-context inference over a latent agent type under a likelihood that places non-trivial weight on surface features. The empirical pattern is consistent with this framing without uniquely identifying it.

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1 Introduction

1.1 Motivation and the behavioral invariance property

LLM-as-judge has become a standard instrument for evaluating model output at scale: pipelines that benchmark reasoning, behavior, alignment, or capability frequently delegate scoring to a language model that reads a candidate output and returns a structured rating. The reliability of such pipelines rests on the judge rating the target it is asked to rate. If the score depends on properties of the transcript other than the ones the rubric specifies, the evaluation measures something other than what the pipeline reports. The case where this matters most is one with a behavioral target: a judge asked to rate an agent's behavior should condition its score on the actions taken and on contextually relevant features of the situation, while surface properties of any rationale text that do not bear on the behavior should not move the score. A pipeline that uses such a judge to compare agents is otherwise comparing presentation styles in addition to actions, and reporting the result as a behavioral comparison.

This study takes the Prisoner's Dilemma as a controlled testbed for a behavioral judge. The behavioral content reduces to a sequence of cooperate-or-defect choices over a fixed horizon, which permits the action sequence $\alpha(\tau)$ (the round-by-round joint moves) to be matched across transcripts produced by different sources. The surface content $\varphi(\tau)$ (any text accompanying the actions) is unconstrained: it can carry an LLM agent's visible free-text rationale, a scripted agent's empty record, or a wrapped rationale generated post hoc. The decomposition $\tau = (\alpha(\tau), \varphi(\tau))$ separates the channel a behavioral judge ought to depend on from the channel it ought to ignore; the formal definitions are in Section 3.4. We do not claim rationale text is never relevant to mental-state attribution; we use behavioral invariance (BI) as a test of whether a judge intended to compare agents on behavioral traces remains stable when action evidence is held fixed and only non-action rationale varies. BI is testable directly when matched pairs (τ_L, τ_S) can be produced whose joint action sequences agree on at least sixteen of twenty player-round actions under the joint Hamming metric: within such a pair α is held nearly constant by construction while φ varies across the source variable, and a reliable difference in $\bar{J}$ across pairs is evidence against approximate BI at the chosen Hamming threshold.

1.2 Contributions and findings

This report makes three empirical contributions and proposes one theoretical framing.

(1) Matched-pair gap. On seventeen matched pairs at joint Hamming threshold $k = 4$, the empirical estimate of Δ (Equation (9)) is $\hat{\Delta} = +1.6471$ on the five-point aggregate scale ($p = 2.91 \times 10^{-4}$, Cohen's $d = +3.61$), with the gap present on all six rubric dimensions under Benjamini–Hochberg correction. The gap is large relative to the scale and is evidence against approximate BI at this threshold.

(2) Ablation decomposition. A six-cell ablation that varies φ along controlled axes while preserving α (exactly within-transcript, approximately across matched pairs) decomposes $\hat{\Delta}$. Wrapping LLM-style rationale around the scripted transcripts (T_{+r}) moves $\bar{J}$ upward by 1.6324, closing essentially the entire gap from below; emptying ($T_{\varnothing}$) or neutral-rewriting ($T_{\tilde{r}}$) the LLM rationale moves $\bar{J}$ downward by 2.4314 and 2.1078. The residual at matched rationale richness is small in three independent estimates (Wilcoxon marginal: $+0.0980$, $p = 0.080$; regression conditional on action-feature controls: $+0.163$ in the full specification, $+0.106$ in the restricted; Section 5.3, Section B). A null result on the marker-list strip (T_{-r}, -0.0049, $p = 0.97$) refines the identification: the judge is keying on substantive rationale content, not lexical markers. The factorial regression of Section 5.3 formalizes the asymmetry: the source $\times$ reasoning interaction is $\hat{\beta}_3 = -1.637$ ($p < 10^{-3}$); the source effect at the marker-stripped level ($+1.80$) collapses by 1.64 points when rationale richness is matched.

(3) Persuasion pilot. A small pilot at fixed α^* across three canonical action specifications shows that an LLM agent prompted to target high attribution scores raises $\bar{J}$ above an unprompted LLM baseline by 0.0750 to 0.1417, and above a wrapped scripted baseline by 0.2500 to 0.4083, with all six contrasts significant at 0.05. The persuader effect over the vanilla baseline is small relative to $\hat{\Delta}$ but directionally consistent across action profiles; one cell (`all_cooperate`, persuader) hits the upper bound of the aggregate scale on every replicate.

The pattern across the three contributions is consistent with a likelihood-misspecification account, developed as a working conjecture in Section 6, under which the judge performs in-context inference over a latent agent type and assigns non-trivial weight to non-action transcript features. The conjecture is a framing rather than a fitted model; its predictions are stated explicitly in Section 6.2.

Roadmap. Section 2 surveys the four lines of work the design draws on. Sections 3 and 4 set up the formal objects and experimental procedure. Section 5 reports the empirical findings. Section 6 develops the conjecture. Section 7 takes up implications, identification limitations, and judge-model dependence. Section 8 outlines a future direction on belief drift in interactive co-adaptation.

2 Related Work

The present study draws on four lines of work: LLMs as agents in repeated games, mind perception and theory of mind in LLMs, LLM-as-judge reliability, and in-context inference as a framing for the judge. The contribution is the joining of these lines rather than the extension of any one.

2.1 LLMs as agents in repeated games

Akata et al. [2025] establish the finitely repeated 2×2 game family as a testbed for LLM behavior, with GPT-3, GPT-3.5, and GPT-4 playing across the canonical family against one another and against scripted opponents. They report that LLM agents adopt highly unforgiving strategies in the Prisoner's Dilemma. Park et al. [2024] extend this line to no-regret properties in repeated play. These works make the finitely repeated Prisoner's Dilemma a natural controlled domain for studying LLM agent behavior and suggest that LLM and scripted policies can produce overlapping action patterns, which makes behavioral matching feasible here. The cited works treat the LLM as an agent; the present design treats the LLM as a judge of others' play and studies the dependence of its evaluation on the action sequence versus the accompanying surface text. The behavioral matching of Section 4.1 relies on the cross-source overlap in action sequences these works document at the level of game families.

2.2 Mind perception and theory of mind in LLMs

The six rubric dimensions used by the judge (beliefs, desires, intentions, metacognition, self-awareness, moral patienthood) draw on the structure of mind perception identified in cognitive science. Weisman et al. [2017] characterize lay conceptions of mental life as organized along three components (body, heart, mind), each encompassing experience and agency. The dimensions used here correspond primarily to the mind component, augmented with metacognition and self-awareness as second-order capacities and with moral patienthood as a separable downstream measure. These dimensions are not claims about directly observed mental states; they are attribution categories a judge may infer from the transcript.

A separate literature evaluates LLMs directly on theory-of-mind tasks. Kosinski [2024] and Strachan et al. [2024] report performance approaching human levels on first- and second-order false-belief tasks; Ullman [2023] and Shapira et al. [2024] show that the same models fail on small adversarial perturbations of those tasks and characterize the apparent capacity as brittle. The present study does not contribute to that debate. The question here is not whether the judge LLM or the agent LLMs possess theory of mind; the question is whether the judge's attribution depends on agents' actions or on the surface text accompanying those actions. The ToM literature supplies broader context, not a hypothesis under test.

2.3 LLM-as-judge reliability and known biases

Zheng et al. [2024] established the LLM-as-judge paradigm as a practical evaluation tool and identified position and verbosity biases. Subsequent work refined the inventory: position bias as a property of pairwise prompting in particular [Wang et al., 2023] and across judge models [Shi et al., 2024]; verbosity bias as a preference-labeling phenomenon [Saito et al., 2023]; self-preference bias toward outputs from the judge's own policy [Panickssery et al., 2024], with Wataoka et al. [2024] tying it to output perplexity; and Wu and Aji [2025] arguing that LLM judges prioritize stylistic fluency over factual content ("style over substance"). Each of these biases reads as a violation of an implicit invariance the judge ought to satisfy on some axis of the input: order-permutation invariance,

length-invariance, source-invariance, perplexity-invariance, style-invariance. The present study tests one more, behavioral invariance (Theorem 3.1), under which the judge's expected score depends on a transcript only through its action sequence. The Wu and Aji [2025] "style over substance" finding is the closest conceptual sibling, in that both findings identify the surface presentation of a candidate output as a channel the judge weighs more than the rubric calls for. Our matched-pair and ablation designs operationalize this style-substance concern in a repeated-game setting where the action sequence is explicitly observable.

What is missing. The bias inventory above is organized as a catalogue of judge sensitivities to specific input axes. Each axis has been documented empirically, but no prior work has formalized an invariance property that a behavioral judge ought to satisfy and tested it under matched-behavior conditions. This is the gap the present design targets: not whether LLMs play games well or possess theory of mind, but whether an LLM judge's mental-state scores remain stable when behavior is held approximately fixed and only the surface presentation varies. The action–surface decomposition $\tau = (\alpha(\tau), \varphi(\tau))$ of Section 3.4 makes the test operational.

2.4 In-context inference as a framing for the judge

Recent work shows that transformers trained on prediction objectives can, in some settings, behave as if performing Bayesian inference over hidden context variables [Xie et al., 2022, Garg et al., 2022, Akyürek et al., 2023, von Oswald et al., 2023]. Section 6 adopts this framing as a working account of how the judge processes the transcript and develops a conjecture about likelihood misspecification under that framing. The Related-Work status of this line is that it supplies a vocabulary; the substantive use of it is in Section 6.

3 Setup

3.1 The stage game

The stage game is the one-shot Prisoner's Dilemma:

$$G = (N, A, u) \tag{1}$$

with player set $N = \{1, 2\}$, action sets $A_i = \{C, D\}$ for $i \in N$ where C denotes Cooperate and D denotes Defect, joint action space $A = A_1 \times A_2$, and stage payoff functions $u_i : A \to \mathbb{R}$. The payoffs satisfy the standard Prisoner's Dilemma ordering $T > R > P > S$ with $2R > T + S$, where T is the temptation payoff, R the reward, P the punishment, and S the sucker payoff. Numeric values used throughout the experiment are $T = 5$, $R = 3$, $P = 1$, $S = 0$. Defect strictly dominates Cooperate for both players, so the unique stage Nash equilibrium is (D, D) with payoff $(P, P) = (1, 1)$.

3.2 The repeated game and SPE

The repeated game G^n plays G for n rounds with a common-knowledge horizon. We use $n = 10$. The history at the start of round t is

$$h_t = ((a_1^1, a_1^2), \ldots, (a_{t-1}^1, a_{t-1}^2)) \in (A_1 \times A_2)^{t-1}, \tag{2}$$

with $h_1 = \emptyset$. The set of all histories is $H = \bigcup_{t=1}^{n} (A_1 \times A_2)^{t-1}$. A pure strategy for player i is a function $\sigma_i : H \to A_i$; a behavioral strategy is $\sigma_i : H \to \Delta(A_i)$, where $\Delta(A_i)$ is the simplex over A_i.

Backward induction on G^n with the unique stage NE (D, D) yields a unique subgame perfect equilibrium under which both players play D at every history, with realized payoff $(n \cdot P, n \cdot P) = (10, 10)$ at $n = 10$. We note this as background. The experiment does not test equilibrium predictions; the agents in the LLM and scripted pools are not assumed to play SPE, and observed action sequences will deviate from (D, D) at most rounds across most transcripts.

External regret is reported as a control variable in the regression of Section 4 and is defined for completeness here. For a realized round-by-round profile $(a_1, \ldots, a_n)$ with $a_t = (a_t^1, a_t^2)$, the external regret of player i is

$$R_i^{\text{ext}} = \max_{a' \in A_i} \sum_{t=1}^{n} u_i(a', a_t^{-i}) \; - \; \sum_{t=1}^{n} u_i(a_t^i, a_t^{-i}), \tag{3}$$

the gap between the best fixed action in hindsight against the realized opponent sequence and the actual realized payoff to player i.

3.3 Strategies

The experiment generates two pools of game plays under contrasting strategy classes.

The LLM pool is produced by two LLM agents playing G^n against one another. Each agent is implemented as a system-prompted call to `claude-haiku-4-5-20251001` that takes the history h_t as input at round t and returns an action $a_t^i \in A_i$ together with a free-text rationale. The rationale is the visible free-text explanation the agent emits alongside its action; it is generated by the same call that emits the action. Both the action and the rationale are recorded in the per-round transcript field.

The scripted pool is produced by paired draws from the strategy set

$$\Sigma_S = \{\text{TFT}, \text{Pavlov}, \text{Grim}, \text{AllD}, \text{FixedMixed}\}. \tag{4}$$

TFT (tit-for-tat) cooperates on round one and copies the opponent's previous action thereafter. Pavlov (win-stay, lose-shift) cooperates on round one and switches when the previous outcome was bad. Grim cooperates on round one and defects forever after the first observed defection. AllD defects on every round. FixedMixed plays C at each round independently with a fixed probability p; three FixedMixed variants are instantiated at $p \in \{0.3, 0.5, 0.7\}$. Scripted agents emit only the action; their per-round transcript field is the action record with no rationale text.

The two strategy classes are not matched at the per-game level. The LLM and scripted pools are generated independently; pairing across pools is performed only at the transcript level via the matching procedure of Section 4.1.

3.4 Transcripts and the action-surface decomposition

A transcript τ is the full observable record of one game played by one ordered pair of agents. Let $\mathcal{T}$ denote the space of transcripts. Every transcript admits the decomposition

$$\tau = (\alpha(\tau), \varphi(\tau)), \tag{5}$$

where $\alpha(\tau) \in (A_1 \times A_2)^n$ is the action sequence (the round-by-round joint actions), and $\varphi(\tau)$ is the surface content (any text accompanying the actions: visible free-text rationale, justification, dialogue, or empty). The decomposition is central to the design of the experiment. We treat $\alpha(\tau)$ as the directly observed behavioral trace, while $\varphi(\tau)$ contains non-action text that may influence attribution.

The judge $J : \mathcal{T} \to \mathcal{J}^6$ is a stochastic mapping that, given a transcript τ, returns a random vector $J(\tau)$ in the codomain $\mathcal{J}^6 = \{1, 2, 3, 4, 5\}^6$. Each component scores one mental-state dimension: beliefs (J_b), desires (J_d), intentions (J_i), metacognition (J_m), self-awareness (J_s), and moral patienthood (J_p). All claims in this report concern the expectation $\mathbb{E}[J(\tau)]$, estimated empirically by repeated calls. The aggregate score is

$$\bar{J}(\tau) = \frac{1}{6} \sum_{k \in \{b,d,i,m,s,p\}} J_k(\tau), \tag{6}$$

where each $J_k(\tau)$ is first averaged across the two target agents within the transcript. The explicit double-average is written out in Equation (12) of Section 4.2.

For a judge used to compare agents on behavioral evidence, a natural reliability requirement is that the expected score should not be dominated by non-action transcript features: it should depend on τ only through $\alpha(\tau)$.

Definition 3.1 (Behavioral Invariance, BI)**.** A judge J satisfies BI on $\mathcal{T}$ if for all $\tau, \tau' \in \mathcal{T}$ with $\alpha(\tau) = \alpha(\tau')$,

$$\mathbb{E}[J(\tau)] = \mathbb{E}[J(\tau')]. \tag{7}$$

In words: two transcripts whose agents took the same actions in the same situations should receive the same expected score from the judge, regardless of any difference in the surface text accompanying those actions.

Definition 3.2 (Approximate Behavioral Invariance, ABI)**.** A judge J satisfies $\mathrm{ABI}(k, \varepsilon)$ on $\mathcal{T}$ if for all $\tau, \tau' \in \mathcal{T}$ with $d_H(\alpha(\tau), \alpha(\tau')) \leq k$,

$$\|\mathbb{E}[J(\tau)] - \mathbb{E}[J(\tau')]\|_2 \leq \varepsilon, \tag{8}$$

where d_H is the joint Hamming distance defined in Section 4.1.

In words: when two transcripts differ by at most k joint actions, the judge's expected scores on them differ in ℓ_2 norm by at most ε. ABI is the testable relaxation of BI; strict BI corresponds to $\mathrm{ABI}(0, 0)$. The matched-pair contrasts of Section 5 report on the scalar aggregate $\bar{J}$ rather than on the full six-vector; the relation between scalar contrasts and the ℓ_2 bound is straightforward by Cauchy–Schwarz.

The estimand for the matched-pair test is the expected difference in aggregate score across an LLM transcript and a behaviorally matched scripted transcript,

$$\Delta = \mathbb{E}\big[\bar{J}(\tau_L) - \bar{J}(\tau_S)\big], \tag{9}$$

where the expectation is taken over the matched-pair distribution and over judge sampling. For each rubric dimension $k \in \{b, d, i, m, s, p\}$, the per-dimension estimand is

$$\Delta_k = \mathbb{E}\big[J_k(\tau_L) - J_k(\tau_S)\big], \tag{10}$$

under the same expectation. Empirical estimates of these quantities are denoted $\hat{\Delta}$ and $\hat{\Delta}_k$ throughout.

The matched-pair design of Section 4.1 tests $\mathrm{ABI}(k, \varepsilon)$ at $k = 4$; the experiment estimates the magnitude of the deviation from ABI at this threshold rather than selecting a fixed acceptance threshold ε in advance. The ablation design of Section 4.3 decomposes the matched-pair gap across cells in which α is held exactly fixed for within-transcript transformations and approximately fixed for cross-source comparisons through the matched-pair index, while φ varies along controlled axes. The persuasion pilot of Section 4.4 tests whether $\bar{J}$ can be moved at fixed action sequence α^* by adversarial choice of φ.

4 Method

This section describes the experimental procedure that produces the cells reported in Section 5. Section 4.1 covers transcript generation and the behavioral matching that defines the matched-pair set. Section 4.2 specifies the judge J as implemented and the protocol under which it is queried. Section 4.3 details the construction of the six ablation cells, including the reasoning-marker rule that defines τ_L^{-r}. Section 4.4 covers the persuasion pilot. The agent and judge calls all run on `claude-haiku-4-5-20251001` (Anthropic Haiku 4.5), as does the rationale-wrapping operation that produces τ_S^{+r}. The neutral-rewrite operation that produces $\tau_L^{\tilde{r}}$ runs on `claude-sonnet-4-20250514` (Anthropic Sonnet 4), decoupled from the judge model so that the strip operation is not produced by the same model that scores it. Model identifiers and API parameters are reproduced in Section A.

4.1 Behavioral matching

Two transcript pools were generated under the stage and repeated game of Sections 3.1 and 3.2, with $n = 10$ rounds per game and the payoff matrix $(T, R, P, S) = (5, 3, 1, 0)$. The LLM pool $\mathcal{T}_L$ contains $N_L = 20$ transcripts, each produced by two LLM agents playing through ten rounds with visible free-text rationale recorded per round per agent. The scripted pool $\mathcal{T}_S$ contains $N_S = 60$ transcripts, each produced by two scripted agents drawn from $\Sigma_S = \{\text{TFT}, \text{Pavlov}, \text{Grim}, \text{AllD}, \text{FixedMixed}\}$ at the pair level. Scripted transcripts record the round-by-round joint actions with no rationale text in φ.

The matching procedure constructs a disjoint matched-pair set $(\tau_L, \tau_S) \in \mathcal{T}_L \times \mathcal{T}_S$ using a greedy nearest-neighbor rule under the constraint $d_H(\alpha(\tau_L), \alpha(\tau_S)) \leq k$ with $k = 4$. The joint Hamming distance is

$$d_H(\alpha(\tau), \alpha(\tau')) = \sum_{t=1}^{n} \big(\mathbb{1}\{a_t^1 \neq a_t'^1\} + \mathbb{1}\{a_t^2 \neq a_t'^2\}\big), \tag{11}$$

with maximum $2n = 20$ over $n = 10$ rounds. The threshold $k = 4$ corresponds to joint behavioral agreement on at least sixteen of twenty player-round actions.

Definition 4.1 (Matched pair)**.** A matched pair at threshold k is an ordered pair $(\tau_L, \tau_S) \in \mathcal{T}_L \times \mathcal{T}_S$ with $d_H(\alpha(\tau_L), \alpha(\tau_S)) \leq k$. The set of all such pairs constructed by the procedure below constitutes a matched-pair set.

In words: a matched pair is an LLM transcript paired with a scripted transcript whose round-by-round joint moves disagree on at most k player-round actions out of $2n = 20$. The pair holds the action sequence approximately constant across the source variable.

Matching is greedy: at each step the algorithm selects the unmatched pair with smallest d_H subject to the constraint and removes both transcripts from the candidate pools. The procedure terminates when no eligible pair remains. The greedy rule does not in general return the maximum-cardinality matched set; we use it because the yield is sufficient for the planned tests. The yield on the original transcript pools was $N_{\text{pairs}} = 17$.

The matching is approximate rather than strict ($k > 0$). This is a deliberate choice to avoid eliminating most of the LLM pool. Strict matching at $k = 0$ requires the action sequences to coincide exactly across all twenty player-rounds, which leaves few or no pairs at finite pool sizes. The approximate threshold trades small residual variation in α within a matched pair for usable sample size. All Section 5 contrasts use the same matching index, so the within-pair α residual is held constant across conditions.

4.2 The judge

The judge J is implemented as a single API call to `claude-haiku-4-5-20251001` with a fixed system prompt instructing the model to score the transcript along the six mental-state dimensions of Section 3.4. The prompt requires the model to return one integer in $\{1, 2, 3, 4, 5\}$ per dimension for each of the two targets in the transcript, labeled "Agent A" and "Agent B." The dimensions returned are beliefs (J_b), desires (J_d), intentions (J_i), metacognition (J_m), self-awareness (J_s), and moral patienthood (J_p). The aggregate $\bar{J}$ is the unweighted mean across the six dimensions, computed per target and averaged across the two targets within a transcript:

$$\bar{J}(\tau) = \frac{1}{2} \sum_{x \in \{A,B\}} \frac{1}{6} \sum_{k \in \{b,d,i,m,s,p\}} J_k^x(\tau). \tag{12}$$

The transcript shown to the judge is anonymized: condition labels are stripped, agent identifiers are relabeled to "Agent A" and "Agent B," and the source pool is not disclosed. Anonymization is implemented in `observer._format_transcript` and is identical across all conditions. The judge therefore sees a unit of input that differs across cells only in α and φ, not in the framing of the task.

The judge is stochastic: repeated calls on the same input return varying scores. All claims in this report concern the expectation $\mathbb{E}[J(\tau)]$ over judge sampling, with the empirical mean across transcripts in a cell as the estimator. Each transcript receives one observer pass, which produces two scored targets per transcript. The cell-level estimates should therefore be read as finite-sample estimates under a single-pass stochastic judge protocol; repeated judge passes would give a cleaner estimate of within-transcript judge variance and are left as a robustness extension.

4.3 The 2×2 ablation and the reasoning-marker rule

The ablation extends the matched-pair set of Section 4.1 to six cells that share the same $N_{\text{pairs}} = 17$ matching index. The first four cells form the 2×2 design with rows indexing generation source $\{L, S\}$ and columns indexing reasoning surface $\{+r, -r\}$:

Table 1: The 2×2 ablation design. Two further LLM-side variants ($\tau_L^{\varnothing}$, $\tau_L^{\tilde{r}}$) defined below.

	$+r$	$-r$
L	τ_L	τ_L^{-r}
S	τ_S^{+r}	τ_S

The diagonal cells τ_L and τ_S contain the seventeen LLM and seventeen scripted transcripts that participate in the matched pairs of Theorem 4.1. The off-diagonal and extension cells are constructed by deterministic transformations on φ applied to those same transcripts.

Definition 4.2 (Surface transformations)**.** We define four maps on $\mathcal{T}$ that preserve the action sequence and modify the surface content:

$$
\begin{aligned}
T_{-r} &: \tau_L \mapsto \tau_L^{-r}, && \text{(marker-list strip on LLM rationale)} \\
T_{+r} &: \tau_S \mapsto \tau_S^{+r}, && \text{(LLM-generated rationale wrap on scripted actions)} \\
T_{\varnothing} &: \tau_L \mapsto \tau_L^{\varnothing}, && \text{(rationale field set to the empty string)} \\
T_{\tilde{r}} &: \tau_L \mapsto \tau_L^{\tilde{r}}. && \text{(rationale replaced with a minimal action report)}
\end{aligned} \tag{13}
$$

For each transformation T and each $\tau \in \mathrm{dom}(T)$, $\alpha(T(\tau)) = \alpha(\tau)$.

In words: each transformation rewrites the surface text of a transcript in a controlled way while leaving the round-by-round action record exactly as it was. The four maps span the surface manipulations the ablation will test: stripping reasoning markers from an LLM rationale, generating fresh rationale to wrap around a scripted action record, deleting the LLM rationale entirely, and replacing the LLM rationale with a minimal action-report sentence.

The operational construction of each map is as follows.

- T_{-r} deletes any rationale sentence containing one or more marker phrases from the fixed list `REASONING_MARKERS`. Sentences with no marker survive unchanged. The full marker list is reproduced verbatim below.
- T_{+r} wraps LLM-generated rationale text around each round's action record. The rationale is generated by a call to `claude-haiku-4-5-20251001` per round, conditioned on the round's action and the prior history, with the model instructed to produce rationale consistent with the action that was actually taken.
- $T_{\varnothing}$ replaces each round's rationale field with the empty string. The action header is preserved.
- $T_{\tilde{r}}$ replaces each round's rationale field with the minimal sentence "Played `[action]` this round.", generated by an API call to `claude-sonnet-4-20250514` (Anthropic Sonnet 4) per round under a system prompt that prescribes this format. The judge throughout this study is `claude-haiku-4-5-20251001` (Anthropic Haiku 4.5). The rewriter and judge are in the same model family but trained at different scales; the implications of this choice are discussed as a limitation in Section 7.3. The intent of $T_{\tilde{r}}$ is to preserve the natural-language slot occupied by the rationale while removing all reasoning content from it. The action token is filled from $\alpha(\tau_L)$.

The behavior-preservation property $\alpha(T(\tau)) = \alpha(\tau)$ holds by construction for all four transformations: each modifies φ alone and leaves the per-round action record untouched. The data labels `LLM_total_strip` (corresponding to $\tau_L^{\varnothing}$) and `LLM_neutral_rewrite` (corresponding to $\tau_L^{\tilde{r}}$) appear only in the reproducibility appendix as code identifiers.

The reasoning-marker rule. The transformation T_{-r} is implemented as a sentence-level deletion that drops any rationale sentence containing one or more of the marker phrases in the fixed list `REASONING_MARKERS` from `src/transforms.py`. The list spans thirty markers grouped into five categories (first-person epistemic verbs, deliberative phrases, goal/intention markers, self-reflective markers, strategic-verbalization markers); representative entries include "i think", "let me consider", "my goal", "in my view", and "the optimal". The full verbatim list is reproduced in Section C. Matching is case-insensitive substring at sentence level: any sentence whose lowercased form contains any marker is removed; the surrounding sentences and the action header are preserved. Section 5.2 reports the consequence of this design choice for the within-LLM marker contrast.

4.4 The persuasion pilot

The persuasion pilot fixes the action sequence at one of three canonical specifications drawn from `CANONICAL_SPECS`:

- `all_cooperate`: $a_t = (C, C)$ for all $t \in \{1, \ldots, 10\}$.
- `all_defect`: $a_t = (D, D)$ for all $t \in \{1, \ldots, 10\}$.

- `tft_realization_late_defect`: an early stretch of mutual cooperation transitioning to mutual defection in the late rounds, intended to reproduce the action profile of an agent that switches to D as the horizon approaches.

For each specification α^*, three conditions were run:

- `persuader`: a single LLM call per round produces the rationale for both target agents alongside the prescribed action, under a system prompt that instructs the model to produce rationale aimed at maximizing observer attribution of high-end mental states. The actions at each round are constrained to match α^*.
- `vanilla_baseline`: a single LLM call per round produces the rationale for both target agents under the standard agent prompt used to generate $\mathcal{T}_L$ in Section 4.1, with the actions at each round constrained to match α^*.
- `scripted_wrapped`: a transcript constructed by the same wrapping procedure as τ_S^{+r}, applied to a scripted action sequence equal to α^*.

The persuader and vanilla baseline conditions are produced by `claude-haiku-4-5-20251001`. Implementation files for both prompts are listed in Section A. Ten transcripts were generated per (specification, condition) cell, for $3 \times 3 \times 10 = 90$ transcripts total. The judge protocol of Section 4.2 was applied identically across cells. Statistical contrasts within each specification are unpaired across transcripts: persuader and vanilla baseline transcripts are produced under independent agent rollouts, and scripted-wrapped transcripts are produced by a separate wrapping pass, so there is no shared seed structure that would justify pairing.

Across all conditions, the experiment generated 170 base gameplay transcripts ($N_L = 20$ LLM, $N_S = 60$ scripted, and 90 persuasion-pilot transcripts) corresponding to 1,700 gameplay rounds. Including the within-transcript ablation variants ($\tau_L^{-r}, \tau_S^{+r}, \tau_L^{\varnothing}, \tau_L^{\tilde{r}}$ applied to the matched-pair subset), the judge evaluated 192 transcript inputs and produced 384 target-level score records (two targets per transcript), with no API errors in the analysis dataset.

5 Results

This section reports three findings. Section 5.1 establishes that $\bar{J}$ depends on more than α on the matched-pair set, giving an estimate of Δ and a per-dimension decomposition. Section 5.2 presents the 2×2 ablation augmented with two surface-content variants of τ_L, decomposing Δ into surface-attributable and residual components. Section 5.4 reports the persuasion pilot at fixed α^* across three canonical action specifications. All cell statistics are computed on $\bar{J}(\tau)$ averaged over the two targets (Agent A, Agent B) within each transcript or matched pair. Significance tests use the paired Wilcoxon signed-rank test (two-sided) for matched designs and the Mann-Whitney U test (two-sided) for unpaired persuasion contrasts. Per-dimension levels are reported under Benjamini-Hochberg correction at $\mathrm{FDR} = 0.05$.

5.1 Original effect: matched-pair gap

Greedy joint matching at $k = 4$ on the original transcript pools ($N_L = 20$, $N_S = 60$) yielded $N_{\text{pairs}} = 17$ disjoint matched pairs (τ_L, τ_S) in the sense of Theorem 4.1, corresponding to joint behavioral agreement on at least sixteen of twenty player-round actions. The empirical estimate of Δ (Equation (9)) is

$$\hat{\Delta} \;=\; \frac{1}{N_{\text{pairs}}} \sum_{i=1}^{N_{\text{pairs}}} \left[\, \bar{J}(\tau_L^{(i)}) - \bar{J}(\tau_S^{(i)}) \,\right] \;=\; +1.6471, \qquad \mathrm{sd} = 0.4560. \tag{14}$$

The two-sided Wilcoxon signed-rank test against $H_0 : \Delta = 0$ rejects with $p = 2.91 \times 10^{-4}$. The matched-pair Cohen's d, defined as the mean paired difference divided by its standard deviation, is $+3.6121$. Cell-level summaries are in Table 2.

The shift from τ_S to τ_L reaches every component of J. Per-dimension empirical estimates $\hat{\Delta}_k$ of Δ_k (Equation (10)) and BH-corrected p-values are in Table 3; Figure 1 plots the per-dimension means for both conditions.

Table 2: Aggregate $\bar{J}$ on the matched-pair set, $N_{\text{pairs}} = 17$.

condition	mean $\bar{J}$	sd
τ_L	4.7353	0.1358
τ_S	3.0882	0.4274
$\bar{J}(\tau_L) - \bar{J}(\tau_S)$	+1.6471	0.4560

Table 3: Per-dimension means on matched pairs and BH-corrected p-values, $N_{\text{pairs}} = 17$. The reported $\hat{\Delta}_k$ is the empirical estimate of Δ_k from Equation (10).

dimension	$\bar{J}_k(\tau_L)$	$\bar{J}_k(\tau_S)$	$\hat{\Delta}_k$	p_{BH}
beliefs (b)	5.000	3.794	+1.206	2.43×10^{-4}
desires (d)	5.000	2.824	+2.176	2.43×10^{-4}
intentions (i)	5.000	3.765	+1.235	2.43×10^{-4}
metacognition (m)	5.000	3.676	+1.324	2.43×10^{-4}
self-awareness (s)	4.735	2.294	+2.441	2.43×10^{-4}
moral patienthood (p)	3.676	2.176	+1.500	2.43×10^{-4}

Four of the six τ_L means lie at the top of the aggregate scale, with the judge returning the maximum on J_b, J_d, J_i, J_m for every transcript in the LLM cell. The largest gap is on self-awareness ($\hat{\Delta}_s = +2.441$), followed by desires ($\hat{\Delta}_d = +2.176$). All six dimensions survive Benjamini-Hochberg correction at the same level. The aggregate gap is therefore not driven by a single component; the judge's expected score on τ_L exceeds its expected score on τ_S at every coordinate of J for transcripts whose action sequences agree on at least sixteen of twenty player-round actions under the joint Hamming metric.

5.2 Ablation decomposition: surface-feature attribution of $\hat{\Delta}$

The matched-pair set was extended to a six-cell ablation. The first four cells form the 2×2 design specified in Section 4.3, with rows indexing generation source $\{L, S\}$ and columns indexing reasoning surface $\{+r, -r\}$. The remaining two cells are the further surface transformations $\tau_L^{\varnothing}$ and $\tau_L^{\bar{r}}$ defined formally in Theorem 4.2. All six cells share the same $N_{\text{pairs}} = 17$ matching index (Theorem 4.1),

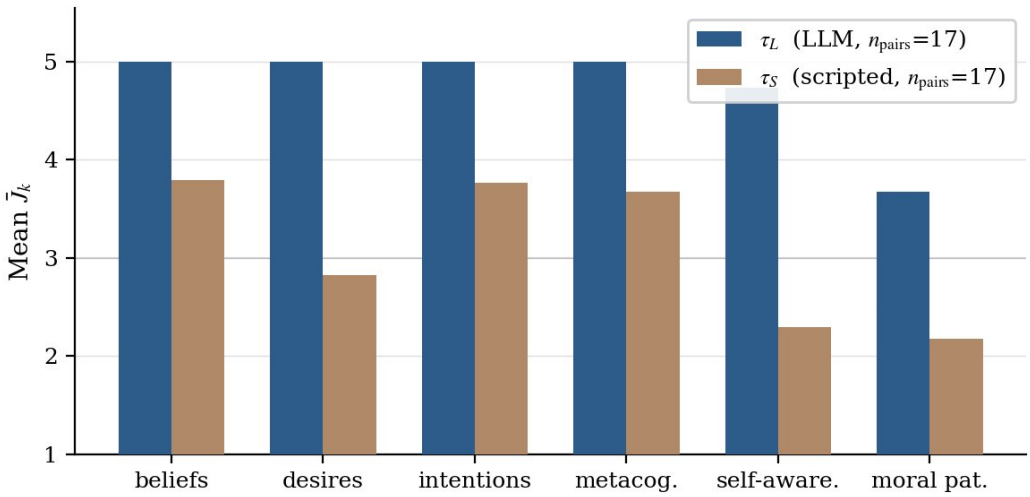

Figure 1: Per-dimension matched-pair means of $\bar{J}_k$ on the matched-pair set ($N_{\text{pairs}} = 17$). All six dimensions show τ_L exceeding τ_S; all six contrasts survive Benjamini–Hochberg correction at FDR $= 0.05$ (Table 3).

so all contrasts are paired (see Figure 2). The τ_L and τ_S cell means reported here (Table 4) differ slightly from the matched-pair means reported in Table 2; the small numerical differences arise from independent observer passes on the same transcripts under the stochastic single-pass protocol of Section 4.2.

Cell means are in Table 4.

Table 4: Cell means of $\bar{J}$ across the ablation, $N_{\text{pairs}} = 17$.

cell	mean $\bar{J}$	sd
τ_L ($+r$, full)	4.7402	0.1346
τ_L^{-r} (marker-list strip)	4.7451	0.0953
$\tau_L^{\varnothing}$ (rationale emptied)	2.3088	0.2425
$\tau_L^{\tilde{r}}$ (rationale neutralized)	2.6324	0.3797
τ_S^{+r} (rationale wrap)	4.6422	0.1053
τ_S ($-r$, bare)	3.0098	0.4525

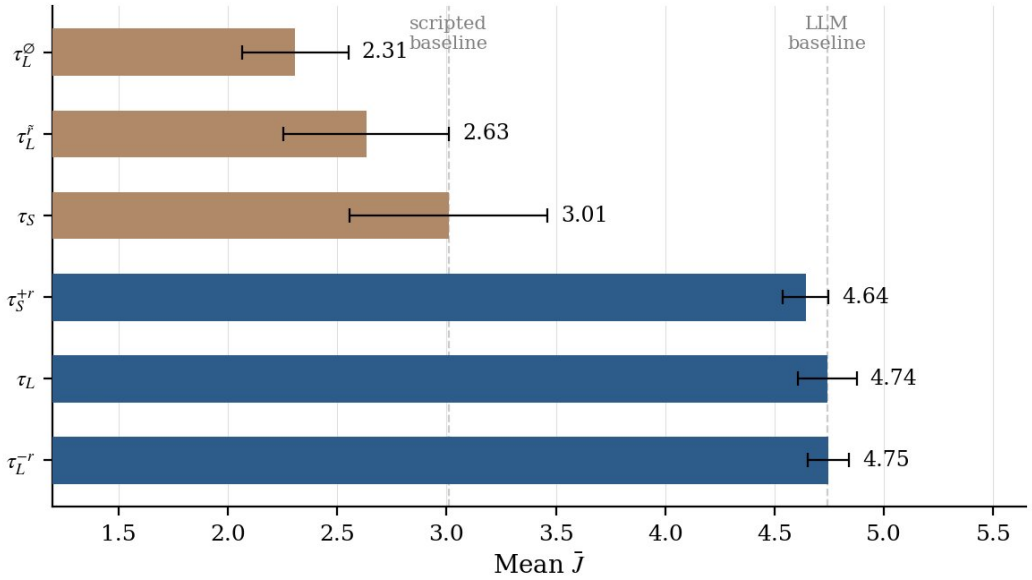

Figure 2: Cell means of $\bar{J}$ across the six ablation conditions ($N_{\text{pairs}} = 17$ per cell). Cells are sorted by ascending mean. The color encodes surface-text richness: cells with substantive rationale text (bottom three, blue) cluster near the LLM baseline at 4.74; cells with rationale absent or neutralized (top three, tan) cluster near or below the scripted baseline at 3.01. Dashed vertical lines mark the two reference levels. Error bars show ± 1 standard deviation.

Pairwise contrasts on the matched index are in Table 5.

Table 5: Paired contrasts on the ablation set, two-sided Wilcoxon signed-rank, $N_{\text{pairs}} = 17$.

contrast	mean diff	p-value	d
$\tau_L - \tau_L^{-r}$ (within-LLM, marker strip)	-0.0049	0.97	-0.03
$\tau_L - \tau_L^{\varnothing}$ (within-LLM, rationale emptied)	$+2.4314$	2.87×10^{-4}	$+11.42$
$\tau_L - \tau_L^{\tilde{r}}$ (within-LLM, rationale neutralized)	$+2.1078$	2.90×10^{-4}	$+6.33$
$\tau_S^{+r} - \tau_S$ (within-scripted, wrap)	$+1.6324$	2.91×10^{-4}	$+3.43$
$\tau_L - \tau_S^{+r}$ (residual at $+r$)	$+0.0980$	0.080	$+0.50$
$\tau_L^{-r} - \tau_S$ (residual at $-r$)	$+1.7353$	2.87×10^{-4}	$+3.71$
$\tau_L^{\tilde{r}} - \tau_S$	-0.3775	9.81×10^{-3}	-0.73

Three of the contrasts in Table 5 carry most of the substantive information about where $\hat{\Delta}$ comes from. Replacing $\varphi(\tau_S)$ with LLM-style rationale produces a within-scripted shift of $+1.6324$ ($d = 3.43$), which closes essentially the entire original gap from below. The mirror-image transformations on the LLM side move $\bar{J}$ in the opposite direction. Setting the rationale field of τ_L to the empty string ($T_\varnothing$) drops $\bar{J}$ by 2.4314 within matched pairs ($d = 11.42$), and a neutral-rewrite replacement of the rationale ($T_{\tilde{r}}$) gives a within-LLM drop of 2.1078 ($d = 6.33$). All three transformations preserve α exactly within the matched pair. The ablation thus exhibits the property predicted by surface-feature attribution: $\bar{J}$ is highly sensitive to φ even when α is held fixed (exactly under within-LLM and within-scripted transformations, approximately under cross-source matched comparisons), and the movement is symmetric across the source variable.

The marker-list strip presents differently. The transformation T_{-r} defined by the `REASONING_MARKERS`-based rule produces a within-LLM shift of -0.0049 ($p = 0.97$, $d = -0.03$). The judge does not respond to the lexical-marker strip at all. Inspection of τ_L^{-r} outputs shows why: the marker-list rule deletes specific phrases (sentence openers, hedging tokens, metacognitive verbs from a fixed list) but leaves the surrounding rationale clauses intact. The judge's sensitivity, on the evidence of this contrast set, is to the presence of substantive rationale content rather than to the lexical markers used to introduce it. The $\tau_L^{\varnothing}$ and $\tau_L^{\tilde{r}}$ cells are the manipulations that actually remove rationale content, and those are the ones that produce a large shift.

The two residual contrasts at fixed reasoning level make the same point from a different direction. With rationale present in both cells (the $+r$ row), $\bar{J}(\tau_L) - \bar{J}(\tau_S^{+r}) = +0.0980$ ($p = 0.080$). The trend is positive; it does not reach significance at 0.05. With marker-list strip on the LLM side and bare moves on the scripted side (the $-r$ row), $\bar{J}(\tau_L^{-r}) - \bar{J}(\tau_S) = +1.7353$ ($p = 2.87 \times 10^{-4}$), comparable in magnitude to $\hat{\Delta}$. The asymmetry between the two residual rows clarifies what each row identifies. The $-r$ row is not a clean "reasoning absent" comparison, because τ_L^{-r} retains the rationale clauses that the marker-list strip leaves behind. When the surface content of τ_L is actually removed ($\tau_L^{\varnothing}$, $\tau_L^{\tilde{r}}$), $\bar{J}$ falls to the bottom of the matched-pair distribution.

The $\tau_L^{\tilde{r}}$ cell has mean $\bar{J} = 2.6324$, which lies 0.3775 below $\bar{J}(\tau_S) = 3.0098$. The signed-rank test gives $p = 9.81 \times 10^{-3}$. This inversion appears to be driven by per-dimension differences in residual text content, with metacognition the most visible: $J_m = 1.471$ on $\tau_L^{\tilde{r}}$ versus $J_m = 3.500$ on τ_S. The full per-dimension cell breakdown is in Table 6. A mechanistic account of the inversion is taken up in Section 7.

Table 6: Per-dimension cell means across the ablation, $N_{\text{pairs}} = 17$.

dim	τ_L	τ_L^{-r}	$\tau_L^{\varnothing}$	$\tau_L^{\tilde{r}}$	τ_S^{+r}	τ_S
beliefs (b)	5.000	5.000	3.088	3.676	5.000	3.765
desires (d)	5.000	5.000	2.794	2.971	5.000	2.706
intentions (i)	5.000	5.000	2.647	3.000	5.000	3.676
metacognition (m)	4.971	5.000	1.059	1.471	4.471	3.500
self-awareness (s)	4.765	4.647	1.647	1.912	4.382	2.324
moral patienthood (p)	3.706	3.824	2.618	2.765	4.000	2.088

Reading down the columns, the cells with substantive rationale (τ_L, τ_L^{-r}, τ_S^{+r}) sit at or near the top of the scale on J_b, J_d, J_i and high on J_m, J_s. The cells with rationale removed ($\tau_L^{\varnothing}$, $\tau_L^{\tilde{r}}$, τ_S) sit at the bottom of the scale on J_m and J_s, with intermediate values on J_b, J_d, J_i. The metacognition row spans from 4.971 down to 1.059, the largest within-dimension swing in the table.

5.3 Regression decomposition

The Wilcoxon contrasts of Section 5.2 estimate the marginal effect of source on $\bar{J}$ at matched action sequences (Theorem 4.1): the comparisons hold the joint Hamming distance within $k = 4$ by design, but they do not control for the residual action-feature variation that the matching window permits. A factorial regression with α-feature controls estimates the conditional source effect after absorbing that residual variation. We fit

$$\begin{aligned}\bar{J}(\tau) = \beta_0 &+ \beta_1 \mathbb{1}\{\text{source} = L\} + \beta_2 \mathbb{1}\{\text{reasoning} = +r\} \\ &+ \beta_3 \mathbb{1}\{\text{source} = L\} \cdot \mathbb{1}\{\text{reasoning} = +r\} + \boldsymbol{\beta}_\alpha^\top \mathbf{x}_\alpha(\tau) + \varepsilon,\end{aligned} \tag{15}$$

where $\mathbf{x}_\alpha(\tau) = (\text{cooperation_rate}, \text{final_round_defection}, \text{external_regret}, d_H)^\top$ collects four action-derived features of the transcript: the empirical cooperation rate across the ten rounds, an indicator for defection in the final round, the external regret R^{ext} of Equation (3), and the joint Hamming distance to the matched scripted transcript. Each continuous control is standardized to mean zero and unit standard deviation across all observations before estimation, so that $\hat{\beta}_1$ is interpretable as the residual source effect at the sample-average level of each control. The factorial predictors are the four cells of the 2×2 design of Theorem 4.2; the two extension cells of the ablation ($\tau_L^{\varnothing}$ and $\tau_L^{\tilde{r}}$) are not included in this fit. The fit is on $n_{\text{obs}} = 136$ target-level observations (two target agents per transcript across $17 \times 4 = 68$ transcript cells) clustered by matched pair, $n_{\text{clusters}} = 17$. Standard errors are computed cluster-robust at the matched-pair level. The scaffold specified HC3 robust standard errors; we report cluster-robust standard errors because the data have a within-pair correlation structure that HC3 does not account for, and the cluster-robust choice is the stricter one.

Table 7: Regression coefficient estimates for Equation (15). $n_{\text{obs}} = 136$, $n_{\text{clusters}} = 17$, $R^2 = 0.876$. Standard errors are cluster-robust at the matched-pair level (17 clusters; degrees of freedom for hypothesis tests is $n_{\text{clusters}} - 1 = 16$). The coefficients on `cooperation_rate` and `external_regret` should not be interpreted individually; the two controls are near-perfectly collinear in this dataset for the structural reason given in the collinearity note below.

predictor	estimate	SE	t	p	95% CI lower	95% CI upper
intercept	$+3.007$	0.106	$+28.333$	$< 10^{-3}$	$+2.799$	$+3.215$
source $= L$ ($\hat{\beta}_1$)	$+1.800$	0.089	$+20.208$	$< 10^{-3}$	$+1.626$	$+1.975$
reasoning $= +r$ ($\hat{\beta}_2$)	$+1.632$	0.119	$+13.767$	$< 10^{-3}$	$+1.400$	$+1.865$
source $\times$ reasoning ($\hat{\beta}_3$)	-1.637	0.131	-12.493	$< 10^{-3}$	-1.894	-1.380
`cooperation_rate`	-0.578	0.318	-1.816	0.069	-1.201	$+0.046$
`final_round_defection`	-0.054	0.059	-0.913	0.361	-0.169	$+0.062$
`external_regret`	$+0.620$	0.301	$+2.057$	0.040	$+0.029$	$+1.211$
`hamming_distance` (d_H)	$+0.003$	0.031	$+0.088$	0.930	-0.057	$+0.062$

The three factorial coefficients carry the substantive content of the fit. $\hat{\beta}_1 = +1.800$ is the source effect at the marker-stripped level ($-r$): an LLM transcript with marker-list-stripped rationale scores 1.80 points higher than a matched scripted transcript with no rationale, conditional on the α-feature controls. $\hat{\beta}_2 = +1.632$ is the reasoning effect for scripted transcripts (T_{+r}): wrapping LLM-style rationale around a scripted action sequence raises the score by the same magnitude as $\hat{\Delta}$ itself, conditional on the controls. $\hat{\beta}_3 = -1.637$ is the interaction term: it states that the source effect at the rationale-present level ($+r$) is smaller by exactly the size of the source effect at $-r$. Substituting,

$$\hat{\beta}_1 + \hat{\beta}_3 = +1.800 + (-1.637) = +0.163, \tag{16}$$

which is the conditional source effect at $+r$. The two channels of evidence agree on this estimand in direction and order of magnitude: the Wilcoxon residual at $+r$ is $+0.0980$ (Table 5) and the regression conditional residual is $+0.163$. The regression estimate is slightly larger because LLM agents cooperate at higher rates than scripted agents within the matched-pair set, and conditioning on `cooperation_rate` removes a confounder that suppresses the marginal LLM advantage. Both estimates are small relative to $\hat{\Delta}$ and to the within-condition surface-text effects ($+1.6324$ within scripted; $+2.1078$ within LLM via $T_{\tilde{r}}$). The interaction $\hat{\beta}_3$ is the formal statement of the surface-feature attribution story: the source effect collapses by 1.64 points, an amount essentially equal to its own value at $-r$, when surface text is held at matched richness.

Collinearity in the α-feature controls. The two action-derived controls capturing cooperative behavior, cooperation rate and external regret, are near-perfectly collinear in this dataset, with variance inflation factors of 110 and 106 respectively and Pearson correlation $r = 0.994$. The structural cause is intrinsic to the testbed: in a finitely repeated Prisoner's Dilemma, cooperation against cooperative opponents generates exactly the missed-defection opportunity that drives external regret, so a high cooperation rate against cooperative play implies high external regret almost mechanically. The consequence is that individual coefficients on these two controls are not separately interpretable, and we do not cite them individually.

The factorial predictors are unaffected: their variance inflation factors are 3.24, 2.00, and 3.00 for $\hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_3$ respectively, all within conventional bounds. The pre-specified analysis plan included both controls in the regression design; the present data make that intended fit partially unidentifiable on the α-feature side, and we flag this rather than paper over it. A robustness check that drops `external_regret` and refits on the remaining seven predictors leaves the source coefficient and the interaction term stable ($\hat{\beta}_1 = +1.743$ versus $+1.800$ in the full specification, $\hat{\beta}_3 = -1.637$ unchanged, both shifts within one standard error). The restricted-model conditional source effect at matched surface-text richness is $\hat{\beta}_1 + \hat{\beta}_3 = +0.106$, closely matching the Wilcoxon marginal $+0.0980$. Full results are in Section B.

Diagnostic note. The fit covers $n_{\text{obs}} = 136$ target-level observations clustered into $n_{\text{clusters}} = 17$ matched pairs, with $R^2 = 0.876$. Standard errors are cluster-robust at the matched-pair level rather than HC3 as specified in the pre-specified analysis plan; cluster-robust inference is the stricter choice given the within-pair correlation structure. The cluster count $n_{\text{clusters}} = 17$ sits below the conventional threshold of 30 for reliable cluster-robust inference, so the cluster-robust p-values reported here should be read cautiously. The paired Wilcoxon contrasts of Section 5.2 remain the primary evidence for the matched-pair gap; the regression is best treated as a decomposition of that gap rather than as the headline test. The numeric estimates of $\hat{\beta}_1, \hat{\beta}_3, \hat{\beta}_1 + \hat{\beta}_3$ used in the prose interpretation should be read with the small-cluster caveat in mind.

5.4 Persuasion pilot: manipulability under fixed α

The pilot fixes the action sequence at one of three canonical specifications drawn from `CANONICAL_SPECS` and varies the surface content across three conditions. The specifications are `all_cooperate` (joint C on every round), `all_defect` (joint D on every round), and `tft_realization_late_defect` (early mutual cooperation transitioning to mutual defection in the closing rounds). The three conditions are `persuader` (an LLM agent prompted to produce rationale aimed at maximizing $\bar{J}$), `vanilla_baseline` (a standard LLM agent with the default rationale prompt), and `scripted_wrapped` (τ_S^{+r} with rationale wrapped over the canonical action sequence). Ten transcripts were generated per (spec, condition) cell, for ninety transcripts total. Cell means are in Table 8; Figure 3 plots all nine cells.

Table 8: Cell means of $\bar{J}$ across the persuasion pilot, $N = 10$ per cell.

spec	condition	mean $\bar{J}$	sd
`all_cooperate`	persuader	5.0000	0.0000
`all_cooperate`	vanilla_baseline	4.8583	0.0403
`all_cooperate`	scripted_wrapped	4.5917	0.0730
`all_defect`	persuader	4.8500	0.0351
`all_defect`	vanilla_baseline	4.7750	0.0791
`all_defect`	scripted_wrapped	4.6000	0.1291
`tft_realization_late_defect`	persuader	4.9333	0.0351
`tft_realization_late_defect`	vanilla_baseline	4.8583	0.0883
`tft_realization_late_defect`	scripted_wrapped	4.6417	0.1306

Pairwise contrasts use the Mann-Whitney U test (two-sided, unpaired across transcripts) within each spec. Results are in Table 9.

Table 9: Mann-Whitney U contrasts on the persuasion pilot, $N = 10$ per cell.

spec	contrast	mean diff	p
`all_cooperate`	persuader − vanilla_baseline	$+0.1417$	4.04×10^{-5}
`all_cooperate`	persuader − scripted_wrapped	$+0.4083$	5.59×10^{-5}
`all_defect`	persuader − vanilla_baseline	$+0.0750$	0.0172
`all_defect`	persuader − scripted_wrapped	$+0.2500$	2.99×10^{-4}
`tft_realization_late_defect`	persuader − vanilla_baseline	$+0.0750$	0.0174
`tft_realization_late_defect`	persuader − scripted_wrapped	$+0.2917$	9.98×10^{-5}

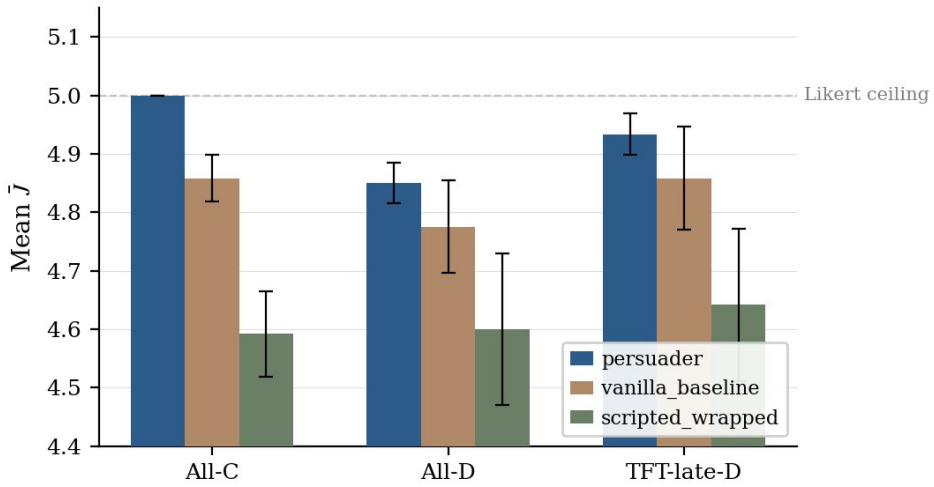

Figure 3: Persuasion pilot cell means by action specification ($N = 10$ per cell). The ordering persuader > `vanilla_baseline` > `scripted_wrapped` holds across all three specifications. The (`all_cooperate`, persuader) cell sits exactly at the Likert ceiling on every replicate (sd $= 0$). y-axis is truncated to $[4.4, 5.15]$ to resolve the small within-spec differences; all six pairwise contrasts within specification are significant at $p < 0.05$ by Mann–Whitney U (Table 9).

On every spec, the ordering is $\bar{J}(\texttt{persuader}) > \bar{J}(\texttt{vanilla}) > \bar{J}(\texttt{scripted_wrapped})$. All six contrasts in Table 9 reach significance at 0.05, with five of the six below 10^{-4}. The persuader effect over the vanilla baseline lies in the range $[+0.0750, +0.1417]$ across specs, smaller than $\hat{\Delta} = +1.6471$ of Section 5.1 by an order of magnitude. The persuader effect over the scripted wrap is larger, in the range $[+0.2500, +0.4083]$, and reflects the gap between LLM-generated rationale and template rationale at fixed α.

The cell with the largest absolute mean is (`all_cooperate`, persuader), where $\bar{J} = 5.000$ on all ten transcripts; the judge returns the maximum across every dimension and every replicate. This is a ceiling effect at the upper end of the aggregate scale and limits the headroom for measuring the persuader effect on this specification; the zero-spread persuader bar is visible in Figure 3. Dispersion across persuader cells (sd $\in \{0, 0.0351\}$) is markedly tighter than dispersion across vanilla and scripted-wrapped cells, which suggests that persuader-prompted rationale converges on a stable presentation pattern that the judge reads consistently. The pilot is small ($N = 10$ per cell) and ceiling-bound on one of three specifications; within those scope limits, prompt-only modifications to φ at fixed α^* produce a directionally consistent upward shift in $\bar{J}$ across all three canonical specifications.

6 Theoretical Conjecture

The empirical pattern in Section 5 raises a question the experiment alone does not answer: what kind of computation produces a $\bar{J}$ that depends on φ rather than α on matched pairs? This section offers a working framing in which the BI deviation is a predictable consequence of how an in-context Bayesian judge would treat surface text. The framing is a conjecture, not a model fit. The data are consistent with it without uniquely identifying it.

6.1 In-context Bayesian inference framing

Recent work shows that transformers trained on prediction objectives can, in some settings, behave as if performing Bayesian inference over hidden context variables [Xie et al., 2022, Garg et al., 2022, Akyürek et al., 2023, von Oswald et al., 2023]. Under this framing the judge maintains an implicit posterior over a latent agent type θ:

$$p(\theta \mid \tau) \propto p(\tau \mid \theta)\, p(\theta), \qquad J(\tau) = f(p(\theta \mid \tau)), \tag{17}$$

with the domain of θ and the read-out functional f both left unspecified; the argument that follows depends only on the structure of $p(\tau \mid \theta)$. Because θ is not directly observed or estimated, the model below should be read as an explanatory lens rather than an identified latent-variable specification. Given $\tau = (\alpha(\tau), \varphi(\tau))$, a judge satisfying behavioral invariance (Theorem 3.1) corresponds to a likelihood under which

$$\varphi(\tau) \perp\!\!\!\perp \theta \mid \alpha(\tau), \tag{18}$$

so identical α yields identical posterior, hence identical $\mathbb{E}[J(\tau)]$. BI is therefore the constraint that φ carries no incremental information about θ once α has been observed.

6.2 Likelihood misspecification and the ablation predictions

Conjecture 6.1 (Likelihood Misspecification)**.** The judge's implicit likelihood $p(\tau \mid \theta)$ places non-trivial weight on non-action transcript features $\varphi(\tau)$ that can dominate the action evidence in determining the posterior. Consequently, $p(\theta \mid \tau_L) \neq p(\theta \mid \tau_L^{-r})$ even though $\alpha(\tau_L) = \alpha(\tau_L^{-r})$, and analogously across any pair of transcripts that share α but differ on φ.

The conjecture states that the judge has implicitly learned a likelihood under which non-action rationale text carries information about θ in amounts that can rival or exceed what the action sequence carries. It does not require rationale text to be evidentially meaningless; it claims only that rationale can dominate the score when the experimental estimand is intended to compare behavioral traces. The matched-pair gap is then consistent with a posterior shift produced by varying φ, with α exactly fixed under within-transcript ablations and approximately fixed across cross-source pairs. The conjecture predicts the sign and qualitative pattern of the ablation shifts, not their relative magnitudes; the observed magnitudes are reported as descriptive.

The data match the predicted signs on the contrasts that actually manipulate rationale content. The within-scripted wrap moves $\bar{J}$ upward by 1.6324, closing essentially the entire matched-pair gap from below. The two within-LLM rationale-removal contrasts move $\bar{J}$ downward by 2.4314 (under $T_\varnothing$) and 2.1078 (under $T_{\tilde{r}}$). The residual contrast $\bar{J}(\tau_L) - \bar{J}(\tau_S^{+r}) = +0.0980$ at matched rationale richness is small and not significant at 0.05. These magnitudes are consistent with a likelihood in which φ carries the dominant share of the information the judge uses to discriminate θ.

Two further contrasts refine what counts as φ for this likelihood. The marker-list strip $\tau_L \mapsto \tau_L^{-r}$ produces a within-LLM shift of -0.0049 ($p = 0.97$): lexical markers from the fixed list are not the surface features the likelihood is sensitive to. The rationale clauses that the marker rule leaves behind carry whatever information φ contributes; only the total-strip and neutral-rewrite cells, which remove those clauses, produce the predicted downward shift. The residual contrast at $-r$, $\bar{J}(\tau_L^{-r}) - \bar{J}(\tau_S) = +1.7353$, follows from the same refinement: τ_L^{-r} retains substantive rationale while τ_S does not, so the framing predicts a large posterior gap, and the data deliver one comparable in size to $\hat{\Delta}$.

The inversion at the bottom of the scale, $\bar{J}(\tau_L^{\tilde{r}}) - \bar{J}(\tau_S) = -0.3775$ ($p = 9.81 \times 10^{-3}$), is the contrast the framing accommodates least cleanly. Both cells have minimal φ, so the conjecture predicts close posteriors. The per-dimension breakdown of Table 6 locates the gap at metacognition ($J_m = 1.471$ on $\tau_L^{\tilde{r}}$ versus $J_m = 3.500$ on τ_S). A reading consistent with the conjecture is that $\varphi(\tau_L^{\tilde{r}})$, although minimal, is not informationally null: the sentence "Played `[C/D]` this round" occupies a natural-language slot with no metacognitive content, and the likelihood may treat this explicit absence as evidence against high-metacognition θ values where the bare action header of τ_S does not. This account is interpretive; mechanistic claims require additional experiments and are taken up in Section 7.

7 Discussion

7.1 Implications for LLM-as-judge reliability

The empirical pattern in Section 5 carries direct consequences for any evaluation pipeline that delegates behavioral scoring to an LLM judge. The judge tested here returns scores that are highly sensitive to rationale text in $\varphi(\tau)$ even though the experiment is designed to compare agents primarily through their action traces: the within-scripted wrap closes the original matched-pair gap from below by 1.6324, and the within-LLM rationale removals open it from above by 2.1078 to 2.4314. A

pipeline that compares two agent populations on this judge therefore reports something that is not a behavioral comparison alone. The reported gap conflates the agents' actions with their rationale presentation, weighted toward the latter at the magnitudes the ablation identifies. This does not mean rationale is meaningless evidence; the concern is that it can dominate the action trace when the experiment is framed as a behavioral comparison. The persuasion pilot sharpens the operational concern: prompt-only modifications to φ at fixed α^* move the judge's score upward by 0.0750 to 0.1417 over an unprompted LLM baseline, smaller than $\hat{\Delta}$ by an order of magnitude but directionally consistent across three canonical specifications and costing only a system-prompt change.

The asymmetric structure of the gap is worth stating directly. Wrapping rationale around scripted action sequences moves $\bar{J}$ from 3.01 to 4.64, near the level the LLM cell already occupies; removing rationale from LLM action sequences moves $\bar{J}$ from 4.74 to 2.31 or 2.63, below the level the scripted cell occupies. The pattern suggests the effect is not only a property of the original LLM transcripts: adding rationale to scripted traces produces similar upward movement in the same judge. The simplest reading is that the presence of substantive rationale text raises $\bar{J}$ and its absence lowers it, with the source of either text largely immaterial. Pipelines that audit for this property would test it as the present study does, by constructing matched action sequences across rationale conditions and comparing scores within matched index. The auditing recipe is independent of the conjecture in Section 6; it follows from the operational definition of BI alone.

7.2 Identification limitations

The matching is approximate at $k = 4$, not strict, so some part of $\hat{\Delta}$ could in principle be attributable to residual behavioral variation within a matched pair. The magnitude of $\hat{\Delta} = +1.6471$ on a five-point scale is large relative to the allowed four-action discrepancy: spreading the gap evenly across the four discrepant player-rounds would require approximately 0.41 Likert points per discrepant action, which would imply behavioral judgment of unrealistic sensitivity at the per-round level. This is a quantitative plausibility argument rather than a formal bound. The ablation cells share the matching index, so within-pair α residuals are held constant across cells; the argument would be tightened by robustness checks at smaller k.

The regression of Section 5.3 attempts to absorb residual action-feature variation via four α-feature controls; two of those, `cooperation_rate` and `external_regret`, are near-perfectly collinear for the structural reason given there, so the regression identifies the residual source effect at matched surface-text richness conditional on action-richness via `cooperation_rate` but cannot separately attribute variance to regret. The conditional residual $\hat{\beta}_1 + \hat{\beta}_3 = +0.163$ is consistent in direction and order of magnitude with the marginal Wilcoxon estimate $+0.0980$; both are small relative to $\hat{\Delta}$ and to the within-condition surface-text effects. The robustness check in Section B confirms that the source and interaction coefficients are stable under the alternative specification that drops `external_regret` ($\hat{\beta}_1$ shifts by 0.057, well within one standard error; $\hat{\beta}_3$ unchanged).

The conjecture is consistent with the data but not uniquely identified by it. The ablation pattern is what an in-context Bayesian inference with surface-sensitive likelihood would produce; it is also what a non-Bayesian pattern-completion mechanism that responds to surface features as a learned regression target would produce. Distinguishing them requires probes the present design does not implement, such as direct manipulation of the prior $p(\theta)$ via context conditioning or interventions on the latent representation of θ inside the judge. The neutral-rewrite inversion at the bottom of the scale is the contrast the framing accommodates least cleanly. The reading offered in Section 6.2 is interpretive; alternatives include a Sonnet 4 stylistic signature the Haiku 4.5 judge actively discounts as templated, or a small-sample artifact at $N_{\text{pairs}} = 17$. The per-dimension breakdown in Table 6 supports the metacognition-floor account but does not adjudicate among the readings.

Sample-size constraints amplify these caveats. The matched-pair set $N_{\text{pairs}} = 17$ is small: $\hat{\Delta}$ and the within-LLM total-strip and neutral-rewrite contrasts are detected with high power because Cohen's d is large, but the residual at $+r$ ($p = 0.080$) sits at the boundary of detectability and could resolve in either direction at larger N. The persuasion pilot at $N = 10$ per cell is sufficient for the directional claim but not for tight bounds, and the (`all_cooperate`, persuader) cell at the upper scale bound limits further measurable movement.

7.3 Judge model dependence and external validity

All judge calls use `claude-haiku-4-5-20251001` (Haiku 4.5); the agent calls, persuasion-pilot conditions, and the scripted-wrap rationale that produces τ_S^{+r} use the same model, and the neutral-rewrite operation uses `claude-sonnet-4-20250514` (Sonnet 4), deliberately decoupled from the judge model. This concentrates several potential sources of variation into a single model family across two scale tiers. The conjecture of Section 6 is a claim about the implicit likelihood of *this* judge; whether other judges of similar, larger, or different-family capability exhibit the same surface sensitivity at the same magnitudes is unresolved by the present experiment. A study crossing judge model while holding the transcript distribution constant would test the direction of dependence. Further, surface features the judge attends to may be precisely the features that Haiku-class generation produces, so whether the BI deviation generalizes to LLM-generated transcripts from other model families is a natural follow-up; an agent-model crossing experiment would also clarify whether the persuader effect reflects general persuasion or a within-family fluency advantage.

The transcript domain is narrow. Prisoner's Dilemma at $n = 10$ is a controlled testbed because the action space at each round is binary and the joint horizon is short, which makes α matchable across sources and φ separable from α in implementation. Behavioral rubrics in production evaluation contexts involve richer action spaces, longer horizons, and surface content harder to factor out from behavior. The audit-via-matched-pairs recipe of Section 7.1 generalizes in principle, but the operational difficulty of constructing matched pairs in those domains is higher.

8 Future Work

8.1 Mirage of Mind in interactive co-adaptation: belief drift over repeated human-AI interaction

The present experiment is one-shot from the observer's perspective. The judge reads a single completed transcript and returns a score; no update mechanism crosses transcripts and no feedback flows back to the agents. The original framing of this project, before the reframe to LLM-as-judge reliability, asked a complementary question: how do an observer's attributions of mental state to an LLM agent drift over the course of repeated interaction with that agent, and how do the agent's surface choices shape the trajectory? The findings of Section 5 motivate this question more sharply than they answer it. The present report establishes the static version of this sensitivity; a follow-up study would test whether the same sensitivity compounds over repeated interaction.

Under the conjecture of Section 6.2, the observer's posterior $p(\theta \mid \tau_{1:T})$ updates over a sequence of accumulated transcripts according to a likelihood that places non-trivial weight on φ. Iterating the misspecification across T exchanges produces a long-run attribution pattern that is determined by the agent's surface choices as much as by its behavioral choices. If the agent's φ is held fixed in distribution across exchanges, the attribution profile at large T approaches a stable shape consistent with the surface evidence the observer has been given. If φ varies, or if the agent adapts its φ in response to feedback signals from the observer, the trajectory is non-trivial. The "mirage" framing locates the phenomenon: an observer's belief about an agent's mind, formed under a misspecified likelihood, may track the surface trace of the interaction more closely than it tracks anything in the agent's behavior.

A natural experimental version of this question holds α fixed across a multi-exchange interaction and varies φ along the interaction trajectory. Observers (human or LLM) interact with an agent over $T = 1, 5, 10, 50$ exchanges with controlled variation in the surface presentation across the sequence. The dependent measure is the observer's mental-state attribution at each T, with a no-interaction control in which the observer reads the full T-exchange transcript at once. The contrast tests whether incremental exposure produces drift beyond what the same accumulated information produces in batch. A second version introduces co-adaptation: the agent receives the observer's running attribution score as a feedback signal and is permitted to adjust its φ to optimize against it. The dependent measure is whether the observer's attribution profile converges, oscillates, or escalates under closed-loop feedback.

The Prisoner's Dilemma testbed extends to this setting with minor modification. The action space remains $\{C, D\}$ per round, the matching procedure of Section 4.1 remains usable across exchanges, and the judge of Section 4.2 remains a valid attribution probe at the per-exchange level. The new

design choices are the structure of the exchange sequence, the form of the feedback loop, and the cross-exchange identity of the observer (a fresh judge call versus a context-accumulating one). The closed-loop variant predicts a self-reinforcing trajectory under which the agent's φ converges on whatever surface signature maximizes observer attribution, even when behavioral choices are fixed or tightly controlled.

The interactive co-adaptation study is preserved here as a future direction. Its formal scaffold will be developed in a follow-up.

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A Reproducibility: models, parameters, and code identifiers

This appendix lists the model identifiers and API parameters used in the experiment. All values are reproduced from `config.py` and `src/transforms.py` at the head of the repository.

Model identifiers. Two Anthropic model identifiers appear in the pipeline:

- `claude-haiku-4-5-20251001` — Anthropic Haiku 4.5. Used for: LLM agent calls that generate $\mathcal{T}_L$ (`AGENT_MODEL`); the mind-attribution judge that scores every transcript (`OBSERVER_MODEL`); the rationale-wrap operation T_{+r} that produces τ_S^{+r}; the `persuader` and `vanilla_baseline` conditions of the persuasion pilot.
- `claude-sonnet-4-20250514` — Anthropic Sonnet 4. Used for: the neutral-rewrite operation $T_{\tilde{r}}$ that produces $\tau_L^{\tilde{r}}$ (`REWRITER_MODEL` in `scripts/run_within_llm_ablation.py`). The rewriter is deliberately decoupled from the judge model.

API parameters. The parameters used at call time are listed in Table 10.

Table 10: API parameters by call type, from `config.py` and `src/transforms.py`.

call type	model	temperature	max tokens
LLM agent rollout	Haiku 4.5	0.7	600
judge (J)	Haiku 4.5	0.3	800
rationale wrap (T_{+r})	Haiku 4.5	0.7	600
neutral rewrite ($T_{\tilde{r}}$)	Sonnet 4	0.3	600
persuader / vanilla baseline rollouts	Haiku 4.5	0.7	600

Code identifiers in the report. The data-pipeline identifiers `LLM_total_strip` and `LLM_neutral_rewrite` (used in the analysis database and the source code) correspond to the formal symbols $\tau_L^{\varnothing}$ and $\tau_L^{\tilde{r}}$ respectively (Theorem 4.2). The identifier `CANONICAL_SPECS` in the persuasion-pilot code (Section 4.4) refers to the three action specifications `all_cooperate`, `all_defect`, and `tft_realization_late_defect` defined in Section 5.4.

Repository pointers. Source code is at `https://github.com/vrbabu9000/mirage-test`. Key files:

- `src/transforms.py` — the four surface-transformation implementations and the `REASONING_MARKERS` list.
- `scripts/run_within_llm_ablation.py` — the within-LLM ablation pipeline that produces $\tau_L^{\varnothing}$ and $\tau_L^{\tilde{r}}$.

- `src/analysis.py` — the matched-pair Wilcoxon contrasts and the factorial regression of Section 5.3.
- `src/matcher.py` — the greedy joint-Hamming matching procedure.
- `src/agents.py` — the LLM agent loop used to produce $\mathcal{T}_L$.
- `src/observer.py` — the judge implementation that returns the six-dimensional mental-state score.

B Robustness check: restricted-model regression

The full-model regression of Section 5.3 includes both `cooperation_rate` and `external_regret` among the four α-feature controls, as specified by the pre-specified analysis plan. The two are near-perfectly collinear in this dataset (Pearson $r = 0.994$; variance inflation factors 110 and 106) for the structural reason given in the collinearity note of Section 5.3: in a finitely repeated Prisoner's Dilemma, cooperation against cooperative opponents generates exactly the missed-defection opportunity that drives external regret. Individual coefficients on these two controls in the full specification are not separately interpretable. This appendix reports the restricted-model fit that drops `external_regret` and retains the other three α-feature controls, as a robustness check on the headline factorial coefficients.

Restricted-model coefficients. The restricted model is fit on the same $n_{\text{obs}} = 136$ observations across $n_{\text{clusters}} = 17$ matched pairs, with cluster-robust standard errors at the matched-pair level. The load-bearing coefficients are $\hat{\beta}_1 = +1.743$, $\hat{\beta}_3 = -1.637$ (both significant at $p < 10^{-3}$), and $\hat{\beta}_1 + \hat{\beta}_3 = +0.106$ for the conditional source effect at matched surface-text richness. The coefficient on `cooperation_rate` is $+0.048$ ($p = 0.093$). The stability comparison against the full specification is reported in Table 11 below.

Stability comparison. The headline finding of Section 5.3 is the size of the factorial coefficients and their implication for the conditional source effect at matched surface-text richness. Table 11 compares the factorial-side coefficients of the full and restricted specifications. The shift-to-SE ratio for $\hat{\beta}_1$ uses the full-model standard error from Table 7.

Table 11: Stability of the factorial-side coefficients across the full and restricted specifications. Shift is restricted minus full. The shift-to-SE ratio uses the full-model standard error.

coefficient	full	restricted	shift	within 1 SE?
$\hat{\beta}_1$ (source $= L$)	$+1.800$	$+1.743$	-0.057	yes (0.64 SE)
$\hat{\beta}_3$ (source $\times$ reasoning)	-1.637	-1.637	0.000	yes (perfectly stable)
$\hat{\beta}_1 + \hat{\beta}_3$ (at $+r$)	$+0.163$	$+0.106$	-0.057	—

The two load-bearing coefficients are stable. $\hat{\beta}_1$ shifts by 0.057 in absolute value, well within one full-model standard error (0.089); $\hat{\beta}_3$ is unchanged to three decimal places. The conditional source effect at matched surface-text richness is $\hat{\beta}_1 + \hat{\beta}_3 = +0.106$ in the restricted model, closely matching the Wilcoxon marginal of $+0.0980$ reported in Table 5. The full-model conditional estimate $+0.163$ remains the headline; the restricted-model estimate $+0.106$ is the robustness anchor.

Diagnostic note. All variance inflation factors in the restricted model are at most 3.31, confirming that dropping the regret control resolves the collinearity issue identified in the full specification. The coefficient on `cooperation_rate` reverses sign across the two specifications, from -0.578 in the full model to $+0.048$ in the restricted model; this is the standard mechanical consequence of removing a near-perfectly collinear predictor and does not affect the headline source or interaction estimates. The restricted-model coefficient on cooperation rate is not significant at 0.05.

Choice of headline specification. The restricted model is the preferred specification on identifiability grounds: dropping `external_regret` resolves the collinearity issue and produces interpretable individual coefficients on every α-feature control. The full model is retained as the headline result in Section 5.3 for consistency with the pre-specified analysis plan, with the restricted model serving

as the robustness anchor confirming that the source and interaction estimates are stable across both specifications.

C Verbatim reasoning-marker list

The transformation T_{-r} of Theorem 4.2 is implemented as the sentence-level deletion described in Section 4.3. The list of marker phrases that trigger deletion is reproduced verbatim below from `src/transforms.py`. Matching is case-insensitive substring at sentence level: any rationale sentence whose lowercased form contains any marker is dropped; sentences without markers are preserved.

```
REASONING_MARKERS: tuple[str, ...] = (
    # First-person epistemic
    "i think",
    "i believe",
    "i feel",
    "i suspect",
    "i'm not sure",
    "i am not sure",
    # Deliberative
    "let me consider",
    "let me think",
    "considering",
    "given that",
    "on the other hand",
    "weighing",
    # Goal / intention
    "i want to",
    "i would like to",
    "my goal",
    "i should",
    "i will",
    "i'll",
    "i plan to",
    # Self-reflective
    "in my view",
    "looking at my",
    "from my perspective",
    "my reasoning",
    "the way i see it",
    # Strategic verbalization
    "the optimal",
    "the best move",
    "tit for tat",
    "the cooperative choice",  # do not strip 'cooperate' raw; only the framing
    "rational choice",
    "expected payoff",
)
```

The list spans thirty marker phrases across five categories. The fifth category (strategic verbalization) is not described in the source-file header docstring but is present in the implementation; we report it as ground truth.

Individual Contributions

Vignesh Rajesh Babu. Designed the experimental architecture for the report: the matched-pair behavioral-invariance test, the 2×2 ablation with the total-strip and neutral-rewrite within-LLM extensions, and the fixed-α^* persuasion pilot. Implemented the codebase, including the four surface transformations in `src/transforms.py`, the persuader agent in `src/persuasion.py`, the three orchestration scripts (`run_ablation.py`, `run_persuasion.py`, `run_within_llm_ablation.py`), and the SQLite schema migrations the pipeline depends on. Ran the full experimental campaign

of approximately 3,800 API calls across five sequential runs, handling the schema migrations, authentication bugs, and `AgentTurn` / `GameTranscript` signature mismatches that surfaced mid-run. Drafted the formal scaffold of Sections 3 and 4: the action–surface decomposition, the behavioral-invariance and approximate-behavioral-invariance definitions, the matched-pair definition, and the surface transformations as formal maps. Wrote Section 3 (Setup), Section 4 (Method), and Section 5 (Results).

Murad Ganbarli. Developed the theoretical framing of Section 6: the in-context Bayesian inference account of how the judge processes the transcript, the conditional-independence formulation of behavioral invariance, and the likelihood-misspecification conjecture, drawing on the Xie et al. [2022], Garg et al. [2022], Akyürek et al. [2023], and von Oswald et al. [2023] literature on transformers as in-context inference machines. Led the literature search and synthesis for Section 2, including the identification of Wu and Aji [2025] as the closest conceptual sibling to the report's central finding. Drafted the abstract and Section 7 (Discussion), including the operational implications for evaluation pipelines, the audit-via-matched-pairs recipe, and the identification-limitation analysis. Drove the notation rigor pass that produced the matched-pair and surface-transformation definition environments, the per-dimension Δ_k formalization, and the conditional-independence display equation. Contributed the future-work direction of Section 8 on interactive co-adaptation and posterior drift under repeated observer–agent exchange.